

A Hybrid Preprocessor Architecture for the Streaming Spectrum Sensor

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Abstract—Previous work introduced the streaming spectrum sensor (StSS), a low-memory field-programmable gate array architecture that serially evaluates trial symbol-rate and center-frequency pairs. This paper describes a fast Fourier transform (FFT) symbol-rate preprocessor intended to reject most trial cycle-frequency bins before streaming spectral correlation analysis. In an illustrative case, reducing 11 downstream evaluations to two gives a 5.5-fold upper bound on the speedup before preprocessor overhead. A two-detector implementation is reported to fit a Virtex-7 XC7VX485T. Because the preserved results contain neither a serial baseline nor post-preprocessor detection, timing-closure, or power measurements, they establish implementation fit rather than measured end-to-end throughput or edge-device suitability.

Index Terms—streaming spectrum sensor, spectral correlation, RF feature detection, FPGA, edge computing

I. INTRODUCTION

Spectrum sensing can provide environmental knowledge that a cognitive radio uses to adapt its operation. One family of feature detectors estimates a signal’s spectral correlation function (SCF), also called cyclic spectral density. Common estimators include the strip spectral correlation analyzer and FFT accumulation method; their resource and resolution tradeoffs are reviewed in [1].

The streaming spectral correlation analyzer (StSCA) was developed to estimate selected SCF points with low block random access memory (BRAM) use [1]. The streaming spectrum sensor (StSS) adds peak detection and a power-scaled threshold to estimate the symbol rate and center-frequency offset of active signals [2]. Related cognitive-networking applications are discussed in [2], [3]. This paper describes an FFT preprocessor and scheduling updates intended to reduce serial StSCA work. The preserved evidence supports resource-utilization claims for one FPGA configu-

ration; claims that require new measurements are identified explicitly.

II. BACKGROUND

Let $X[n]$ be a complex sample sequence acquired at sample rate f_s , let f_k be a trial frequency shift, and let α_v be a trial cycle frequency. For an N -sample record, the finite-record StSCA estimate can be written as

$$y_{v,k}[n] = \sum_{\ell} h_v[\ell] X[n - \ell] e^{-j2\pi f_k(n-\ell)/f_s}, \quad (1)$$

$$\hat{S}_X^{\alpha_v}(f_k) = \frac{1}{N} \sum_{n=0}^{N-1} |y_{v,k}[n]|^2 e^{-j2\pi \alpha_v n/f_s}. \quad (2)$$

Here, $h_v[\ell]$ is the low-pass filter selected for cycle-frequency branch v . Equation (2) estimates one configured (α_v, f_k) point rather than the continuous SCF; its derivation and mapping to symbol rate and center frequency are given in [1].

The center-frequency detector (CFD) applies the test

$$|\hat{S}_X^{\alpha_v}(f_k)|^2 \underset{H_0}{\overset{H_1}{\geq}} \beta_{\text{CFD}} \left(\frac{1}{N} \sum_{n=0}^{N-1} |X[n]|^2 \right)^2. \quad (3)$$

The dimensionless multiplier β_{CFD} sets an operating point; it is not itself the probability of false alarm. The calibration in [2] was specific to that paper’s waveform, channel, search grid, and decision objective. The reported validation scope is M-ary quadrature-amplitude modulation (M-QAM) with square-root-raised-cosine (SRRC) pulse shaping. Operation on other modulation and pulse-shape families was not demonstrated.

III. DESIGN OVERVIEW

A. Symbol-Rate Preprocessor

The original StSCA avoids FFT cores by processing one configured (α_v, f_k) point in a streaming fashion. The preprocessor concept—an FFT of the instantaneous power followed by candidate selection—was described by Koch and Downey [4] and recommended for the StSCA in [1]. The present architecture integrates that concept with the StSS so

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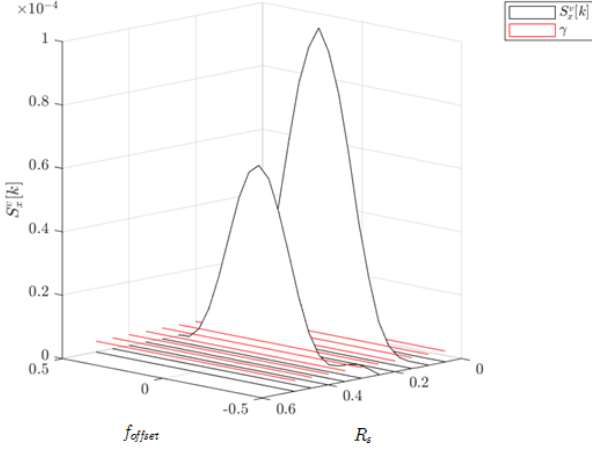


Fig. 1. Illustrative SCF result for two simulated in-band signals: ($R_s = 0.10$ MBd, $f_{\text{offset}} = -0.20$ MHz) and ($R_s = 0.25$ MBd, $f_{\text{offset}} = 0.20$ MHz). The threshold test is applied to the squared-magnitude statistic in (3).

that only selected cycle-frequency candidates are sent to the CFDs.

Figure 1 illustrates 11 evaluated cycle-frequency bins, of which two exceed the plotted threshold. Skipping nine bins reduces downstream evaluations by 81.8%. If each serial CFD evaluation costs T_{CFD} and the preprocessor costs T_{SRP} , the corresponding ideal work ratio is $11T_{\text{CFD}}/(T_{\text{SRP}} + 2T_{\text{CFD}}) < 5.5$. Thus, 5.5 is an upper bound for this example, not a measured throughput gain.

For the SRP, the cyclic autocorrelation estimate at lag $\tau = 0$ is a DFT of the real-valued instantaneous power:

$$\hat{R}_X^{\alpha_m}[0] = \frac{1}{N} \sum_{n=0}^{N-1} |X[n]|^2 e^{-j2\pi mn/N}, \quad \alpha_m = \frac{mf_s}{N}. \quad (4)$$

The DC bin is excluded. Because $|X[n]|^2$ is real, only the unique positive-frequency bins $m = 1, \dots, \lfloor N/2 \rfloor$ need be tested. Off-grid cycle frequencies can leak into adjacent bins, so candidate grouping and the family-wise false-alarm rate across the search grid must be characterized. The SRP and CFD statistics also require separate threshold calibration unless their scaling and noise bandwidths are normalized.

An N -point FFT requires $O(N)$ frame state, but whether that state uses BRAM, distributed memory, or both depends on the selected streaming or burst architecture, scaling, and output order [5]. A single FFT can use less memory than FFT-array SCA architectures, but that comparison was not measured here.

B. Center-Frequency Estimator

In prior simulation, a fixed two-tap smoother was reported as adequate for the tested cases [1]; this observation does not establish periodogram convergence generally. Hardware observations motivated the selectable low-pass-filter bank in

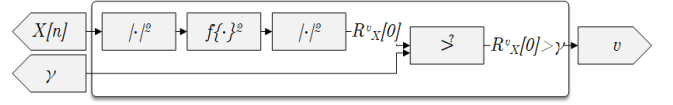


Fig. 2. SRP block diagram. The FFT computes all N DFT bins for the instantaneous-power sequence; the earlier StSCA evaluates configured DFT bins serially.

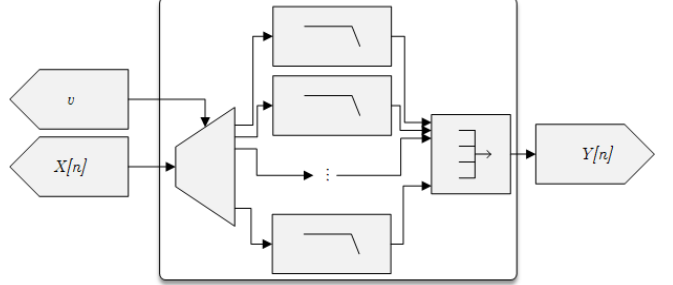


Fig. 3. Selectable low-pass-filter bank. Branch v selects a filter with cutoff $f_{c,v}$; the coefficients, cutoff convention, gain, and equivalent noise bandwidth are required to reproduce the detector calibration.

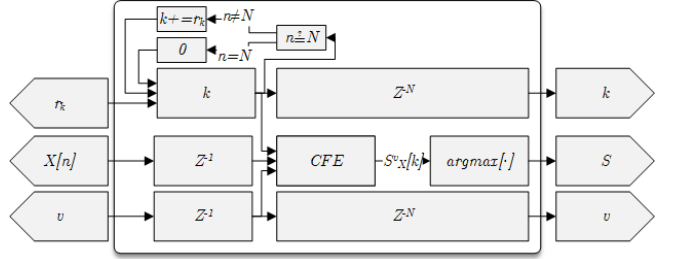


Fig. 4. CFD block diagram. The trial center frequency f_k advances by step r_k ; for a selected α_v , peak selection operates on $|\hat{S}_X^{\alpha_v}(f_k)|^2$.

Fig. 3. The archived record does not retain filter coefficients or a before-and-after detection metric, so the magnitude of the reported accuracy improvement cannot be independently checked.

C. Center-Frequency Detector

The CFD retains the functionality in [2], while thresholding is moved to the top-level StSS and SRP; Fig. 4 shows the simplified data path.

D. Symbol-Rate Distributor

The symbol-rate distributor (SRD) replaces the earlier polling distributor and FIFO commutator [2] with a one-hot priority encoder and decoder. This simplifies the described control path, but the preserved results do not quantify resource savings or reliability. As Fig. 5 notes, a candidate is discarded when every CFD is busy; throughput and detection claims therefore require a candidate-drop measurement.

E. Streaming Spectrum Sensor

The top-level StSS contains the threshold logic, SRP, SRD, and result FIFOs and tracks sample and result counts.

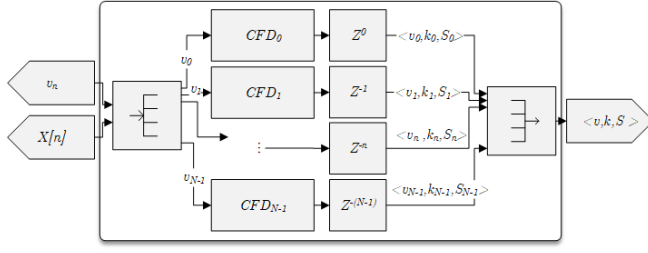


Fig. 5. SRD block diagram. A candidate from the SRP is assigned to a ready CFD; if none is ready, the candidate is discarded. Each CFD starts at the beginning of an N -sample frame; index-based staggering permits result commutation under the assumed fixed-latency, no-stall schedule.

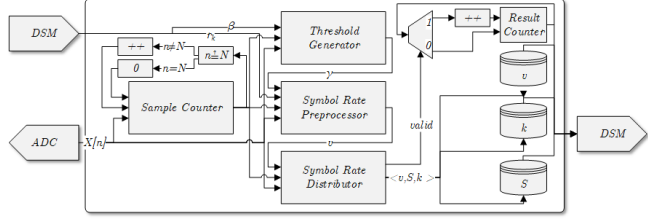


Fig. 6. StSS block diagram. The DSM configures threshold multipliers and center-frequency search resolution. A target false-alarm probability must be mapped to each multiplier by calibration for the selected waveform, channel, grid, filters, and fixed-point scaling.

TABLE I
REPORTED XC7VX485T RESOURCE UTILIZATION

Resource	Used	Available	Utilization
Slice LUTs	22,463	303,600	7.40%
Flip-flops	15,426	607,200	2.54%
36-Kb BRAM equiv.	186.5	1,030	18.11%
DSP48E1 slices	284	2,800	10.14%

It interfaces with an analog-to-digital converter and a CPU-based dynamic spectrum manager (DSM).

Additional CFDs increase concurrent service capacity until another component becomes the bottleneck; they do not reduce the statistical precision required for threshold calibration.

IV. RESULTS AND LIMITATIONS

The resource totals in Table I identify the target as a Virtex-7 XC7VX485T [6]; the original identifier was xc7k7-t, which is not a valid identifiable device. The package and speed grade are not preserved. The reported configuration used 16-bit data, $N = 2^{16}$ samples, and two CFD instances. A latency of 34.36 s from transmission start

to first result was reported, but the sample rate, FPGA clock, search-grid dimensions, FFT configuration, cycle count, and stall behavior are unavailable. Consequently, that number cannot be reproduced or converted into an end-to-end throughput comparison.

The four percentages are arithmetically consistent with the reported counts. The fractional BRAM value denotes an 18-Kb block counted as half of a 36-Kb equivalent. One utilization point demonstrates fit on this device but cannot establish scaling with CFD count, timing closure, power, reliability, or suitability for smaller edge FPGAs.

V. CONCLUSION

The architecture combines an instantaneous-power FFT preprocessor with the StSCA and a two-CFD distributor. The preserved synthesis summary shows that this configuration fits an XC7VX485T. It does not contain the measurements needed to demonstrate a throughput increase or maintain a calibrated detection operating point after preprocessing. A reproducible evaluation should release the HDL and test-bench; identify the complete FPGA part, tool version, clock and FFT settings; report baseline-versus-hybrid frames or candidates per second; measure candidate drops and P_d/P_{fa} over the full search grid; and sweep CFD count to establish resource scaling. Goertzel single-bin DFT implementations and normalized spectral-coherence statistics remain hypotheses to benchmark rather than demonstrated improvements.

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